



Đề mới MAE101 FE B2 FA15

Discrete Mathematics (Trường Đại học FPT)



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MULTIPLE CHOICES QUESTIONS:

QN=1 (9472)	(See picture) [file:9472.jpg] Which of the following are <u>even</u> functions ? (i) $f(x) = x^3 - 1010$ (ii) $f(x) = \sin(2x^4)$ (iii) $f(x) = \sin(x - 10x^5)$ (iv) $f(x) = x + \frac{5}{x} + 2$ (v) $f(x) = \sin x \cdot \cos x$
a.	(i)
b.	(ii)
c.	(iii)
d.	(iv)
e.	(v)
f.	None of the other choices is correct
ANSWER:	B
MARK:	1
UNIT:	1.1
MIX CHOICES:	Yes

QN=2 (9493)	From the graph of $y = f(x-3)+4$, how do we obtain the graph of $y = f(x)$?
a.	Shift down by 4, and then shift to the right 3
b.	Shift up by 4, and then shift to the right 3
c.	Shift down by 4, and then shift to the left 3
d.	Shift up by 4, and then shift to the left 3

e.	None of the other choices is correct
ANSWER:	C
MARK:	1
UNIT:	1.2
MIX CHOICES:	Yes

QN=3 (9503)	(See picture) [file:9503.jpg] If $1 \leq f(x) \leq x^2 + 8x + 8$ for all x , find $\lim_{x \rightarrow -1} f(x)$.
a.	5
b.	-1
c.	0
d.	1
e.	8
f.	None of the other choices is correct
ANSWER:	D
MARK:	1
UNIT:	1.4
MIX CHOICES:	Yes

QN=4 (9543)	(See picture) [file:9543.jpg] Find the constant c that makes g continuous on $(-\infty; \infty)$ $g(x) = \begin{cases} x^2 + c^2 & \text{if } x < 5 \\ cx + 21 & \text{if } x \geq 5 \end{cases}$
a.	1 and 4
b.	0 and 4
c.	0 and 1
d.	1 and;5
e.	5 and 4
f.	None of the other choices is correct
ANSWER:	A
MARK:	1
UNIT:	1.5
MIX CHOICES:	Yes

QN=5 (9545)	(See picture) [file:9545.jpg] Find the limit $\lim_{x \rightarrow \infty} [\sqrt{x+4} - \sqrt{x}]$
a.	0
b.	2
c.	infinity
d.	1
e.	4
f.	None of the other choices is correct
ANSWER:	A
MARK:	1
UNIT:	1.6
MIX CHOICES:	Yes

QN=6 (9571)	(See picture) [file:9571.jpg]
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	Simplify the quotient $\frac{f(x+h)-f(x)}{h}$ for $f(x) = -\frac{1}{x}$ (i) $\frac{-1}{(x+h)x}$ (ii) $\frac{1}{x^2}$ (iii) $\frac{1}{(x+h)x}$ (iv) $\frac{-1}{x^2}$
a.	(i)
b.	(ii)
c.	(iii)
d.	(iv)
e.	None of the other choices is correct
ANSWER:	C
MARK:	1
UNIT:	2.2
MIX CHOICES:	Yes

QN=7 (9584)	(See picture) [file:9584.jpg] A particle moves along a straight line with displacement given by $s(t) = t^2 - 8t + 18$. What is the instantaneous velocity when $t = 4$?
a.	All of the other choices are correct
b.	2
c.	0
d.	4
e.	8

ANSWER:	C
MARK:	1
UNIT:	2.3
MIX CHOICES:	Yes

QN=8 (9626)	(See picture) [file:9626.jpg] Find $y'(1)$ for $y = \frac{x^2 + 2x - 8}{3x + 9}$
a.	61/144
b.	33/72
c.	63/144
d.	31/72
e.	None of the other choices is correct
ANSWER:	C
MARK:	1
UNIT:	2.4
MIX CHOICES:	Yes

QN=9 (9656)	(See picture) [file:9656.jpg] A table of values for f, g, f', g' is given <table><tr><td>x</td><td>$f(x)$</td><td>$g(x)$</td><td>$f'(x)$</td><td>$g'(x)$</td></tr><tr><td>0</td><td>1</td><td>1</td><td>2</td><td>-2</td></tr><tr><td>1</td><td>0</td><td>2</td><td>3</td><td>-1</td></tr><tr><td>2</td><td>4</td><td>0</td><td>5</td><td>6</td></tr></table> Find $h'(2)$ if $h(x) = f(g(x))$.	x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$	0	1	1	2	-2	1	0	2	3	-1	2	4	0	5	6
x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$																	
0	1	1	2	-2																	
1	0	2	3	-1																	
2	4	0	5	6																	
a.	-2																				
b.	0																				
c.	-5																				
d.	12																				
e.	-6																				
f.	None of the other choices is correct																				
ANSWER:	D																				
MARK:	1																				

UNIT:	2.5
MIX CHOICES:	Yes

QN=10 (9678)	(17236) [file:9678.jpg] Differentiate implicitly to find the slope of the curve at the given point $y^3 + yx^2 + x^2 - 3y^2 = 0, (-1, 1)$
a.	-1/2
b.	-2
c.	None of the other choices is correct
d.	3/2
e.	-1
ANSWER:	B
MARK:	1
UNIT:	2.6
MIX CHOICES:	Yes

QN=11 (9722)	(See picture) [file:9722.jpg] Let $y = ax + b$ be the linear approximation for $f(x) = x^{2/3}$ at $x = 1$. Find b.
a.	1/3
b.	-1/3
c.	2/3
d.	- 2/3
e.	None of the other choices is correct
ANSWER:	A
MARK:	1
UNIT:	2.8
MIX CHOICES:	Yes

QN=12 (9733)	(See picture) [file:9733.jpg] Find the absolute maximum and absolute minimum values of $f(x) = x^4 - 32x^2 + 2$ on $[-5,5]$.
a.	None of the other choices is correct
b.	Absolute maximum: 2; absolute minimum: -254
c.	Absolute maximum: 2; absolute minimum: -173
d.	Absolute maximum: 2; absolute minimum: 0
ANSWER:	B
MARK:	1
UNIT:	3.1
MIX CHOICES:	Yes

QN=13 (9774)	(See picture) [file:9774.jpg] Determine where the function $f(x) = (x + 3)^{2/3}$ is concave up and where it is concave down. (i) Concave down on $(-\infty, -3)$ and concave up on $(-3, \infty)$ (ii) Concave down on $(-\infty, -3)$ and $(-3, \infty)$ (iii) Concave up on $(-\infty, -3)$ and $(-3, \infty)$ (iv) Concave up on $(-\infty, -3)$ and concave down on $(-3, \infty)$
a.	(i)
b.	(ii)

c.	(iii)
d.	(iv)
e.	None of the other choices is correct
ANSWER:	B
MARK:	1
UNIT:	3.3
MIX CHOICES:	Yes

QN=14 (9841)	Find the minimum value of the product of two numbers in which one number is 2 more than three times the other.
a.	-1/3
b.	None of the other choices is correct
c.	1
d.	1/3
e.	1/2
ANSWER:	A
MARK:	1
UNIT:	3.5
MIX CHOICES:	Yes

QN=15 (9848)	(See picture) [file:9848.jpg] Use Newton's method with the specified initial approximation x_1 to find x_3, the third approximation to the root of the given equation: $x^5 + 2 = 0, \quad x_1 = -1$
a.	-1.1529
b.	-1.3229
c.	None of the other choices is correct
d.	-2.3058
e.	-2.0027
ANSWER:	A
MARK:	1
UNIT:	3.6
MIX CHOICES:	Yes

QN=16 (9864)	(See picture) [file:9864.jpg]
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	Let $f(x)$ be such that $f(0) = 2$ and $f'(x) = 5x^2 - 7x + 4$ Find $f(1)$.
a.	None of the other choices is correct
b.	25/3
c.	5
d.	4
e.	25/6
ANSWER:	E
MARK:	1
UNIT:	3.7
MIX CHOICES:	Yes

QN=17 (9884)	(See picture) [file:9884.jpg] Approximate the area under the graph of $f(x) = x^2 + 2$ over the interval $[0, 5]$ using 5 subintervals and left endpoints.
a.	40
b.	65
c.	73
d.	66
e.	None of the other choices is correct
ANSWER:	A
MARK:	1
UNIT:	4.1
MIX CHOICES:	Yes

QN=18 (9905)	(See picture) [file:9905.jpg]
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	Let $f(x)$ be a positive continuous function that satisfies $\int_2^b f(z)dz = \int_2^8 f(z)dz + \int_8^{11} f(z)dz$ Find b.
a.	8
b.	19
c.	13
d.	11
e.	None of the other choices is correct
ANSWER:	D
MARK:	1
UNIT:	4.2
MIX CHOICES:	Yes

QN=19 (9941)	(See picture) [file:9941.jpg] Find the derivative of the function $g(x) = \int_x^{x^3} \sin t dt$ (i) $3x^2 \sin x^3 - \sin x$ (ii) $3x^2 \sin x^3 + \sin x$ (iii) $\sin x^3 - \sin x$ (iv) $3x^2 \cos x^3 - \cos x$
a.	(i)
b.	(ii)

c.	(iii)
d.	(iv)
e.	None of the other choices is correct
ANSWER:	A
MARK:	1
UNIT:	4.4
MIX CHOICES:	Yes

QN=20 (9992)	(See picture) [file:9992.jpg] Evaluate the integral $\int e^{-x}(x^2 + 1)dx$ (i) $-e^{-x}(x^2 - 2x + 3) + C$ (ii) $e^{-x}(x^2 + 2x + 3) + C$ (iii) $-e^{-x}(x^2 + 2x + 3) + C$ (iv) $e^{-x}(x^2 - 2x + 3) + C$
a.	(i)
b.	(ii)
c.	(iii)
d.	(iv)
e.	None of the other choices is correct
ANSWER:	C
MARK:	1
UNIT:	6.1
MIX CHOICES:	Yes

QN=21 (10002)	(See picture) [file:10002.jpg]
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Use the Right – endpoint rule with $n = 4$ to estimate the value of the integral

$$\int_1^3 f(x) dx$$

x	1	1.5	2	2.5	3
f(x)	0.31	0.54	0.36	1.35	2.04

a.	2.145
b.	4.290
c.	None of the other choices is correct
d.	3.240
e.	1.620
ANSWER:	A
MARK:	1
UNIT:	6.5
MIX CHOICES:	Yes

QN=22 (10016)	(See picture) [file:10016.jpg]
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	Which of the following integrals is divergent ? (i) $\int_1^{\infty} \frac{2010}{5x^2} dx$ (ii) $\int_1^{\infty} \frac{1}{\sqrt{x^3}} dx$ (iii) $\int_1^{\infty} \frac{1 + xe^{-2x}}{x} dx$
a.	(i)
b.	(ii)
c.	(iii)
d.	None of the other choices is correct
ANSWER:	C
MARK:	1
UNIT:	6.6
MIX CHOICES:	Yes

QN=23 (10087)	(See picture) [file:10087.jpg] A sequence $\{a_n\}$ is defined by $a_1=0, a_{n+1}=2/(a_n+1)$ for $n>0$. Assuming that $\{a_n\}$ is convergent, find its limit.
a.	1
b.	-2

c.	1 and -2
d.	0
e.	None of the others.
ANSWER:	A
MARK:	1
UNIT:	8.1
MIX CHOICES:	Yes

QN=24 (10105)	(See picture) [file:10105.jpg] Investigate the convergence of the series (i) $\sum_{n=0}^{\infty} \left(\frac{n}{n+1} \right)^2$ (ii) $\sum_{n=0}^{\infty} \frac{(-3)^n}{e^{n+2}}$
a.	Both diverge
b.	(i) diverges, (ii) converges
c.	(i) converges, (ii) diverges
d.	Both converge
ANSWER:	A
MARK:	1
UNIT:	8.2
MIX CHOICES:	Yes

QN=25 (10167)	(See picture) [file:10167.jpg] Determine if the series converges conditionally, absolutely or diverges $\sum_{n=1}^{\infty} (-1)^n \frac{1}{n^6 + 1}$
a.	converges conditionally
b.	converges absolutely

c.	diverges
ANSWER:	B
MARK:	1
UNIT:	8.4
MIX CHOICES:	Yes

QN=26 (10183)	(See picture) [file:10183.jpg] Find the radius of convergence of the series $\sum_{n=1}^{\infty} \frac{n^2 x^n}{3^n}$
a.	R =3
b.	R =2
c.	R =1/3
d.	R =1/2
e.	None of the other choices is correct
ANSWER:	A
MARK:	1
UNIT:	8.5
MIX CHOICES:	Yes

QN=27 (10195)	(See picture) [file:10195.jpg]
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	Find the Taylor polynomial T_2 centered at $a = 0$ for the function $f(x) = \cos(e^x)$ (i) $\sin(1) - \cos(1)x - \frac{\sin(1) + \cos(1)}{2}x^2$ (ii) $\cos(1) - \sin(1)x - \frac{\cos(1) + \sin(1)}{2}x^2$ (iii) $\cos(1) + \sin(1)x + \frac{\cos(1) + \sin(1)}{2}x^2$ (iv) $\cos(1) + \sin(1)x - \frac{\cos(1) + \sin(1)}{2}x^2$
a.	(i)
b.	(ii)
c.	(iii)
d.	(iv)
ANSWER:	B
MARK:	1
UNIT:	8.7
MIX CHOICES:	Yes
QN=28 (10203)	(See picture) [file:10203.jpg]

	Which of the following matrices is in reduced row-echelon form? (i) $\begin{bmatrix} 1 & 0 & 0 & 0 & -3 \\ 0 & 0 & 1 & 0 & 4 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix}$ (ii) $\begin{bmatrix} 0 & 1 & 0 & 0 & 5 \\ 0 & 0 & 1 & 0 & -4 \\ 0 & 0 & 0 & -1 & 3 \end{bmatrix}$ (iii) $\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & -2 \end{bmatrix}$
a.	(iii) only
b.	(i) and (ii) only
c.	(ii) and (iii) only
d.	(i) only
e.	None of the other choices is correct
ANSWER:	D
MARK:	1
UNIT:	LA1.1 - Echelon
MIX CHOICES:	Yes

QN=29 (10219)	(See picture) [file:10219.jpg] Find all solutions of the following system of linear equations $\begin{cases} x - y - z = 3 \\ -x - y + z = -1 \end{cases}$
a.	$x = t - 2, y = -1, z = t$
b.	$x = t, y = -1, z = t - 2$
c.	None of the other choices is correct

d.	$x = 8, y = 4, z = 1$
ANSWER:	B
MARK:	1
UNIT:	LA1.1
MIX CHOICES:	Yes

QN=30 (10248)	(See picture) [file:10248.jpg] Solve for z from the system $\begin{cases} x - y + 2z + 3w = 2 \\ x - y + 3z + 2w = 5 \\ 2x - 2y + 4z + 7w = 5 \end{cases}$
a.	4
b.	0
c.	All of the other choices are incorrect
d.	2
e.	1
ANSWER:	A
MARK:	1
UNIT:	LA1.2 - Solve
MIX CHOICES:	Yes

QN=31 (10273)	(See picture) [file:10273.jpg] Find all values of a for which the following homogeneous system $\begin{cases} x + 2y - z = 0 \\ x + 3y + 3z = 0 \\ 2x + 5y + az = 0 \end{cases}$ has only the trivial solution?
a.	All numbers different from -2

b.	2
c.	None of the other choices is correct
d.	All numbers different from 2
e.	-2
ANSWER:	D
MARK:	1
UNIT:	LA1.3
MIX CHOICES:	Yes

QN=32 (10291)	How many solutions would a HOMOGENEOUS system of linear equations of 4 equations and in 4 variables have?
a.	No solution
b.	Unique solution
c.	Infinitely many solutions
d.	There is not enough information
ANSWER:	D
MARK:	1
UNIT:	LA1 - Theory
MIX CHOICES:	Yes

QN=33 (10300)	Let A be a square matrix. Which of the following statements are true? (i) If A is symmetric then $2A+3I$ is also symmetric. (ii) If $2A+3I$ is symmetric then A is also symmetric.
a.	Both (i) and (ii)
b.	Only (i)
c.	Only (ii)
d.	Neither (i) or (ii)
ANSWER:	A
MARK:	1
UNIT:	LA2.1
MIX CHOICES:	Yes

QN=34 (10320)	(See picture) [file:10320.jpg] Let A be an arbitrary square matrix. Which of the following matrices are symmetric: (i) $A + A^T$ (ii) $A + 2A^T$
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a.	(i)
b.	(ii)
c.	(i) and (ii)
d.	None of the other choices is correct
ANSWER:	A
MARK:	1
UNIT:	LA2.2 - Theory
MIX CHOICES:	Yes

QN=35 (10337)	(See picture) [file:10337.jpg] Find the (2, 3)-entry of the product $\begin{bmatrix} -1 \\ 2 \\ 5 \end{bmatrix} \begin{bmatrix} 5 & 3 & -4 \end{bmatrix}$
a.	-8
b.	-15
c.	15
d.	8
e.	-19
f.	None of the other choices is correct
ANSWER:	A
MARK:	1
UNIT:	LA2.2 - Computation
MIX CHOICES:	Yes

QN=36 (10354)	(See picture) [file:10354.jpg]
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	Let A be an $n \times n$ matrix that satisfies $A^3 - 3A^2 + I = 0.$ Choose correct statement: (i) $A^{-1} = 3I - A^2$ (ii) $A^{-1} = A^2 - 3I$ (iii) $A^{-1} = 3A - A^2$ (iv) $A^{-1} = A^2 - 3A$ (v) A is not invertible
a.	(i)
b.	(ii)
c.	(iii)
d.	(iv)
e.	None of the other choices is correct
f.	(v)
ANSWER:	C
MARK:	1
UNIT:	LA2.3 - Theory
MIX CHOICES:	Yes

QN=37 (10374)	(See picture) [file:10374.jpg] If $A = \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$, then the (2, 1)-entry of the matrix $AB^T - A^{-1}B$ is:
a.	13/2
b.	-5/2
c.	3/2
d.	-11/2

e.	None of the other choices is correct
ANSWER:	A
MARK:	1
UNIT:	LA2.3 - Computation
MIX CHOICES:	Yes

QN=38 (10380)	(See picture) [file:10380.jpg] Let $T: R^2 \rightarrow R^2$ be the linear transformation such that $T \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -3 \\ 6 \end{bmatrix}, \quad T \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \\ -6 \end{bmatrix}.$ Compute $T \begin{bmatrix} 7 \\ 3 \end{bmatrix}$. (i) $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ (ii) $\begin{bmatrix} 1 \\ -2 \end{bmatrix}$ (iii) $\begin{bmatrix} 4 \\ -9 \end{bmatrix}$ (iv) $\begin{bmatrix} -1 \\ 7 \end{bmatrix}$
a.	(i)
b.	(ii)
c.	(iii)
d.	(iv)
e.	None of the other choices is correct
ANSWER:	B
MARK:	1
UNIT:	LA2.5 - Linear trans
MIX CHOICES:	Yes

QN=39 (10416)	Suppose that a 5×5 matrix A can be carried to the identity matrix by using exactly three row operations, in the following order: (1) Add a multiple of 3 of the first row to the second row (2) Interchange the second and the third row (3) Dividing the third row by 4. Find the determinant of A.
a.	-4

b.	1024
c.	4
d.	All of the other choices are incorrect
e.	-1/4
ANSWER:	A
MARK:	1
UNIT:	LA3.1 - Theory
MIX CHOICES:	Yes

QN=40 (10426)	(See picture) [file:10426.jpg]
$\text{Let } \begin{vmatrix} a & m & d \\ b & n & e \\ c & p & f \end{vmatrix} = 10. \text{ Find } \begin{vmatrix} 2a+3d & d & -m \\ 2b+3e & e & -n \\ 2c+3f & f & -p \end{vmatrix}.$	
a.	-20
b.	-60
c.	20
d.	60
e.	None of the other choices is correct
ANSWER:	C
MARK:	1
UNIT:	LA3.1 - Computation
MIX CHOICES:	Yes

QN=41 (10450)	If A is a 3 x 3 matrix and $\det(A) = 2$, find $\det(\text{adj}(A))$.
a.	4
b.	3
c.	2
d.	8
e.	None of the others.
ANSWER:	A
MARK:	1
UNIT:	LA3.2 - Theory
MIX CHOICES:	Yes

QN=42 (10463)	(See picture) [file:10463.jpg]
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	Let $A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$. Find the (3,1)-entry of the adjugate of A.
a.	dh-eg
b.	eg-dh
c.	bf-ce
d.	cd-bf
ANSWER:	A
MARK:	1
UNIT:	LA3.2 - Computation
MIX CHOICES:	Yes

QN=43 (10469)	(See picture) [file:10469.jpg] Find the eigenvalues of the matrix $\begin{bmatrix} 2 & 0 & 1 \\ 1 & 1 & 1 \\ -12 & 11 & 4 \end{bmatrix}$
a.	3; 3; -1
b.	3; 3; 1
c.	3; -1; -1
d.	3; -1; 1
e.	None of the other choices is correct
ANSWER:	B
MARK:	1
UNIT:	LA3.3 - Eigenvalues
MIX CHOICES:	Yes

QN=44 (10485)	(153) (See picture) [file:10485.jpg]
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	Given that -3 is an eigenvalue for the matrix $\begin{bmatrix} 3 & -6 \\ 8 & -11 \end{bmatrix}$ Find all eigenvectors corresponding to this eigenvalue $\lambda = -3$.
a.	$t(1, -1)$, t is nonzero
b.	$t(1, -2)$, t is nonzero
c.	$t(2, 1)$, t is nonzero
d.	$t(1, 1)$, t is nonzero
e.	None of the other choices is correct
ANSWER:	D
MARK:	1
UNIT:	LA3.3 - Eigenvectors
MIX CHOICES:	Yes

QN=45 (10518)	Let $U = \text{span}\{(1, 1, 2, 1), (0, 1, 1, -2)\}$. Find all values of t such that $(1, t, 3, 4)$ is in U .
a.	There is no such t
b.	-2
c.	All nonzero numbers
d.	None of the other choices is correct
e.	All number different from -1
ANSWER:	A
MARK:	1
UNIT:	LA5.1 - Spanning
MIX CHOICES:	Yes

QN=46 (10537)	(See picture) [file:10537.jpg]
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	Which of the following subsets are independent in $\mathbb{R}^4$? (i) $\{[1 \ 2 \ 3 \ 4]^T, [2 \ 0 \ 1 \ -1]^T, [1 \ -1 \ 0 \ 3]^T\}$ (ii) $\{[2 \ 0 \ 1 \ -1]^T, [1 \ 2 \ -1 \ 1]^T, [3 \ 2 \ 0 \ 0]^T\}$
a.	(i)
b.	(ii)
c.	(i) and (ii)
d.	None of the other choices is correct
ANSWER:	A
MARK:	1
UNIT:	LA5.2 - Independence
MIX CHOICES:	Yes

QN=47 (10558)	Let u, v be linearly independent vectors in $\mathbb{R}^5$. Let $U = \text{span}\{u, v, 2u + 3v, u - v\}$. Find $\dim U$.
a.	There is not enough information
b.	3
c.	None of the other choices is correct
d.	5
e.	2
ANSWER:	E
MARK:	1
UNIT:	LA5.2 - Dimension
MIX CHOICES:	Yes

QN=48 (10572)	(See picture) [file:10572.jpg]
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	Let X and Y be vectors in $\mathbb{R}^n$. Which of the following statement are true? (i) If $\{X+Y, X-Y\}$ is an orthogonal set then $X = Y$ (ii) If $\{X+Y, X-Y\}$ is an orthogonal set then $X = -Y$ (iii) If $\{X+Y, X-Y\}$ is an orthogonal set then $\ X\ = \ Y\$
a.	(iii) only
b.	(i) and (ii) only
c.	All of (i), (ii) and (iii)
d.	None of the other choices is correct.
ANSWER:	A
MARK:	1
UNIT:	LA5.3
MIX CHOICES:	Yes

QN=49 (10596)	(See picture) [file:10596.jpg] Find the dimension of the column space of $\begin{bmatrix} -1 & 7 & 0 & 3 & 1 \\ 1 & -1 & 0 & -1 & -1 \\ 0 & -3 & 0 & -1 & -1 \\ 0 & 5 & 3 & 4 & -3 \end{bmatrix}$
a.	3
b.	1
c.	2
d.	5
e.	4
ANSWER:	E
MARK:	1
UNIT:	LA5.4 - Computation
MIX CHOICES:	Yes

QN=50 (9912)	(See picture) [file:9912.jpg] Evaluate $\int_1^4 \frac{t^7 - t^3}{t^5} dt$
a.	259/12
b.	79/4
c.	275/12
d.	81/4
e.	None of the other choices is correct
ANSWER:	D
MARK:	1
UNIT:	4.3
MIX CHOICES:	Yes

READING QUESTIONS:
FILL BLANK QUESTIONS:
MATCH QUESTIONS: